

Predictive Forecasting of Care Load & Placement Demand

Time-Series and Machine Learning Approaches for Forecasting HHS Care Load and Discharge Demand

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Abstract

This project presents a predictive forecasting framework for estimating future care load and placement-related demand within the U.S. Department of Health and Human Services (HHS) care system. The study uses daily operational data containing information on children apprehended and placed in CBP custody, children in CBP custody, children transferred out of CBP custody, children in HHS Care, and children discharged from HHS Care.

The original dataset contained 1,170 rows and six variables. After removing blank records, 720 recorded daily observations were identified between January 12, 2023 and December 21, 2025. The recorded dates contained substantial gaps, so the data was converted into a continuous daily time series and missing numerical observations were interpolated.

The methodology combines baseline, statistical, and machine-learning forecasting approaches. Naive Persistence and 7-day Moving Average models were used as baseline methods. Exponential Smoothing and ARIMA were evaluated as statistical forecasting approaches. Random Forest and Gradient Boosting Regressors were developed using lag features, rolling statistics, flow signals, and calendar features. A separate machine-learning model was developed for short-term discharge demand forecasting.

The final forecasting system generates 30-day forecasts for HHS Care load and discharge demand. Approximate residual-based prediction intervals are provided to communicate forecast uncertainty. An intake-versus-exit scenario analysis is also included. The results are presented through an interactive Streamlit dashboard containing forecast visualizations, model comparison, scenario analysis, and relative-load monitoring.

The system is designed as a decision-support and planning tool rather than a replacement for operational judgment.

Keywords: Time Series Forecasting, HHS Care, Machine Learning, Random Forest, Gradient Boosting, ARIMA, Discharge Demand, Forecasting, Streamlit, Predictive Analytics

1. Introduction

Organizations responsible for child welfare and placement operations must continuously respond to changes in the number of children entering care, remaining in care, and exiting care. Changes in these flows can affect operational planning, placement requirements, staffing needs, and resource allocation.

The dataset used in this project contains daily information related to children in U.S. Department of Health and Human Services care. Historical reporting provides information about previous conditions but does not directly provide estimates of future care load or discharge demand.

Predictive forecasting can transform historical observations into forward-looking information. Time-series models identify temporal patterns, while machine-learning models can incorporate lagged observations, rolling statistics, flow signals, and calendar information.

This project therefore develops a forecasting framework combining traditional time-series methods with machine-learning techniques. The system forecasts HHS Care load, short-term discharge demand, intake-versus-exit scenarios, approximate forecast uncertainty, and relative historical load conditions.

2. Problem Statement

The available HHS-related operational dataset contains daily historical observations but does not itself provide a predictive mechanism for estimating future care load and discharge demand.

Research question: How can historical HHS care and flow data be transformed into a forecasting system capable of estimating short-term care load and discharge demand while communicating uncertainty and supporting operational scenario analysis?

3. Objectives

- Forecast the future number of children in HHS Care.
- Develop a separate forecasting model for children discharged from HHS Care.
- Compare assumed intake levels with predicted discharge demand.
- Compare baseline, statistical, and machine-learning forecasting approaches.

- Provide approximate prediction intervals using historical residual variability.
- Develop an interactive Streamlit dashboard for forecasts, comparisons, scenarios, and monitoring.
- Provide forward-looking information as an analytical input for planning.

4. Dataset Description

Column	Description
Date	Date of observation
Children apprehended and placed in CBP custody	Number of children apprehended and placed in CBP custody
Children in CBP custody	Number of children in CBP custody
Children transferred out of CBP custody	Number transferred out of CBP custody
Children in HHS Care	Number of children in HHS Care
Children discharged from HHS Care	Number discharged from HHS Care

The raw dataset contained 1,170 rows and six columns, including 450 completely blank rows. After cleaning, 720 recorded observations remained between January 12, 2023 and December 21, 2025. The expected number of calendar days was 1,075, leaving 355 missing dates.

5. Data Preprocessing

5.1 Removal of Blank Records

The 450 completely blank rows at the end of the file were removed.

5.2 Date Conversion

The Date column was converted to datetime and used as the time-series index.

5.3 Missing Dates

The 1,075 expected calendar dates were compared with 720 recorded dates, identifying 355 missing dates.

5.4 Daily Continuity

The series was reindexed over the complete date range and missing numerical observations were interpolated. A `was_missing` indicator was retained. No numerical missing values remained after preprocessing.

6. Exploratory Data Analysis

6.1 HHS Care Trend

The HHS Care series was plotted over time to identify long-term changes and short-term fluctuations. The latest recorded value was 2,484 children on December 21, 2025.

6.2 Correlation Analysis

Variable Pair	Correlation
HHS Care ↔ HHS Discharges	0.93
HHS Care ↔ Transfers	0.71
HHS Care ↔ Apprehensions	0.70
HHS Care ↔ CBP Custody	0.66
Apprehensions ↔ CBP Custody	0.96

These are descriptive correlations and do not establish causation.

6.3 Time-Series Decomposition

STL decomposition with a weekly period of seven separated observed, trend, seasonal, and residual components. Weekly patterns were visible, particularly in earlier portions of the dataset, while seasonal amplitude appeared lower during

portions of the later period.

7. Feature Engineering

- **Lag features:** lag_1, lag_7, and lag_14.
- **Rolling means:** 7-day and 14-day rolling means using shifted historical observations.
- **Rolling variance:** 7-day and 14-day rolling variance.
- **Flow signal:** Transfers – Discharges, with a lagged version used for future forecasting.
- **Calendar features:** day of week and month.

8. Modeling Methodology

Baseline models: Naive Persistence and 7-Day Moving Average.

Statistical models: Exponential Smoothing and ARIMA.

Machine-learning models: Random Forest Regressor and Gradient Boosting Regressor.

A separate modeling pipeline was created for discharge demand.

9. Train-Test Strategy

Random train-test sampling was avoided because this is a time-series problem. A chronological 80/20 split was used. The final HHS Care forecasting feature setup used 848 training observations and 212 testing observations. The training period covered approximately January 26, 2023 to May 22, 2025, with testing extending through December 21, 2025.

10. Evaluation Metrics

10.1 Mean Absolute Error

$MAE = (1/n) \sum |y_{\text{actual}} - y_{\text{predicted}}|$. It measures average absolute prediction error.

10.2 Root Mean Squared Error

$RMSE = \sqrt{(1/n) \sum (y_{\text{actual}} - y_{\text{predicted}})^2}$. It gives greater weight to larger errors.

10.3 Mean Absolute Percentage Error

MAPE expresses error as a percentage. It can be unstable when actual discharge values are very small or zero, so MAE and RMSE are important for discharge interpretation.

11. HHS Care Model Results

Model	MAE	RMSE	MAPE
Naive Persistence	6.65	8.25	0.30%
7-Day Moving Average	21.56	26.58	0.96%
Exponential Smoothing	211.56	264.82	9.29%
ARIMA (1,1,1)	233.32	286.88	11.00%
Random Forest	39.29	61.26	1.86%
Gradient Boosting	54.25	76.37	2.57%

Random Forest achieved MAE 39.29, RMSE 61.26, and MAPE 1.86%. Gradient Boosting achieved MAE 54.25, RMSE 76.37, and MAPE 2.57%. The Naive Persistence baseline had lower error in this particular evaluation setup, showing why simple baselines are important. Some baseline/statistical and machine-learning evaluations used different forecasting procedures; standardized rolling-origin evaluation is a future improvement.

12. Discharge Demand Forecasting

A separate model was created for Children discharged from HHS Care using lag 1, lag 7, lag 14, 7-day and 14-day rolling means, day of week, and month.

Model	MAE	RMSE	MAPE
Random Forest	3.47	4.67	30.79%
Gradient Boosting	4.43	5.45	54.67%

MAPE is relatively high because small and zero discharge values can make percentage errors disproportionately large. MAE and RMSE are therefore more useful for practical interpretation.

13. Future HHS Care Forecast

The final forecasting system generated a 30-day forecast from December 22, 2025 through January 20, 2026.

Date	Predicted HHS Care
Dec 22, 2025	2,471.97
Dec 23, 2025	2,471.77
Dec 24, 2025	2,471.97
Dec 25, 2025	2,472.20
Dec 26, 2025	2,472.06

Horizon	Average Predicted HHS Care
7 days	2,470.07
14 days	2,473.96
30 days	2,480.30

The forecast remained relatively stable over the 30-day horizon and reflects the assumptions used in recursive forecasting.

14. Future Discharge Demand Forecast

Date	Predicted Discharge Demand
Dec 22, 2025	11.69
Dec 23, 2025	11.87
Dec 24, 2025	11.93
Dec 25, 2025	11.65
Dec 26, 2025	10.78

The average predicted discharge demand over the 30-day period was approximately 9.00 discharges per day.

15. Forecast Uncertainty

Forecast	Residual Std. Dev.	Approx. Interval Width
HHS Care	≈55.75	≈109.28
Discharge	≈4.62	≈9.06

These are residual-based approximate prediction intervals, not formal statistical confidence intervals.

16. Scenario Analysis

The recent 30-day average transfer level was approximately 9.44 transfers per day. Low, expected, and high intake scenarios were defined as 7.56, 9.44, and 11.33 respectively. Net Imbalance = Assumed Intake – Predicted Discharge.

Scenario	Assumed Intake	Avg. Daily Imbalance	30-Day Cumulative
Low Intake	7.56	–1.44	–43.27
Expected Intake	9.44	+0.45	+13.40

Scenario	Assumed Intake	Avg. Daily Imbalance	30-Day Cumulative
High Intake	11.33	+2.34	+70.07

These are sensitivity-analysis assumptions, not direct predictions of future transfers.

17. Relative-Load Monitoring

The dataset contains no explicit operational capacity variable. A historical relative-load reference was therefore created using the 90th percentile of HHS Care, approximately 9,771 children. The latest observed value was 2,484. This percentile is not an actual operational capacity limit.

18. Streamlit Dashboard

- Forecast horizon selector: 7, 14, and 30 days.
- HHS Care model selector: Random Forest and Gradient Boosting.
- HHS Care forecast with approximate uncertainty band.
- Separate discharge-demand forecast.
- Model comparison using MAE, RMSE, and MAPE.
- Intake-versus-exit scenario analysis.
- Historical relative-load monitoring.

19. System Architecture

Raw Dataset → Data Cleaning → Date Conversion & Missing-Date Detection → Continuous Daily Time Series → EDA → Feature Engineering → Time-Based Train/Test Split → Baseline, Statistical and ML Models → Model Evaluation → Future Forecasting → Uncertainty Estimation → Scenario Analysis → Streamlit Dashboard.

20. Recommendations

- Use forecasts as analytical inputs alongside operational knowledge rather than automatic decisions.
- Monitor recent observations and forecast updates regularly.
- Maintain simple baselines such as Naive Persistence for benchmarking.
- Continue modeling discharge demand separately from overall HHS Care load.
- Incorporate actual facility or placement capacity if it becomes available.
- Use rolling-origin or walk-forward validation in future versions.
- Retrain or recalibrate models as new observations become available.

21. Limitations

- The original dataset contained 355 missing calendar dates; interpolation introduces estimated values.
- The available data covers approximately three years and may not capture all long-term patterns.
- No actual capacity variable is available, so the relative-load indicator is not an actual capacity-stress measure.
- Prediction intervals are residual-based approximations.
- Recursive future forecasts require assumptions about future feature values.
- Discharge MAPE can be unstable for small or zero actual values.
- Some model evaluations used different forecasting procedures; unified rolling-origin evaluation would strengthen comparison.

22. Future Scope

- Add holidays, seasonal events, policy changes, facility capacity, geographic information, placement availability, and other operational indicators.
- Evaluate SARIMA, Prophet, XGBoost, LightGBM, LSTM, and Temporal Fusion Transformers.
- Implement automated model retraining.
- Use probabilistic forecasting for calibrated prediction intervals.
- Connect the dashboard to an automated data source.
- Add explicit capacity-based early warning when actual capacity data becomes available.

23. Conclusion

This project developed an end-to-end predictive forecasting framework for HHS care load and discharge demand using historical daily operational data.

The workflow included data cleaning, exploratory analysis, continuous time-series construction, lag and rolling features, flow signals, calendar features, time-based evaluation, statistical models, and machine-learning models.

The final system produces forecasts up to 30 days, approximate residual-based uncertainty intervals, and intake-versus-exit scenario analysis through an interactive Streamlit dashboard.

The framework demonstrates how time-series analysis and machine learning can support forward-looking operational analysis while highlighting the importance of baseline models, careful validation, uncertainty communication, and recognition of data limitations.

24. References

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Appendix A — Key Project Results

Metric	Value
Raw rows	1,170
Blank rows removed	450
Recorded observations	720
Expected calendar days	1,075
Missing dates	355
Final continuous period	Jan 12, 2023 – Dec 21, 2025
Latest HHS Care	2,484
30-day average predicted HHS Care	≈2,480
30-day average predicted discharge	≈9.00/day

Metric	Value
Forecast period	Dec 22, 2025 – Jan 20, 2026

Appendix B — Project Deliverables

- Research Paper
- Executive Summary
- Google Colab Notebook
- Streamlit Dashboard
- Forecast Data Files
- Model Evaluation Results
- Scenario Analysis
- GitHub Repository
- README
- Requirements File
- Deployed Dashboard
- Feedback/Demonstration Video